How Do Web-Pretrained DINOv3 Features Extend to Martian Terrain? Appearance, Adaptation and Errors on AI4Mars

Authors to be added
Affiliation to be added

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Abstract

We ask how the features of a DINOv3 ViT-B/16 pretrained on web images extend to Martian rover imagery (AI4Mars: soil, bedrock, sand, `big_rock`), from three angles: **P1 data** (is the gap one of image appearance?), **P2 model** (where is it worth adapting the backbone?) and **P3 errors** (where does it fail, and does it know?). A 4.6 k-parameter linear probe on the frozen backbone reaches 0.674 / 0.723 mIoU on the expert test sets M1 / M2, statistically indistinguishable from a DeepLabV3-R50 trained end to end with ∼8,600× more parameters (0.667 / 0.713; one under-trained seed). Changing how the images look (histogram equalisation, CLAHE, standardisation, blur) does not move the probe, and rank-8 LoRA on blocks 1–2, 6–7 or 11–12 adds only +0.005 to +0.007 with no difference between zones. What does separate the results is label ambiguity (mIoU 0.862 where three experts agree vs. 0.546 where one labelled) and `big_rock`: calibration is good overall (ECE 0.029) but confidence exceeds accuracy by 0.67 on pixels predicted `big_rock`. Of five hypotheses registered before the runs, two are supported and three inconclusive; we report them as such.

1 Introduction

Self-supervised backbones such as DINOv2 and DINOv3 [9, 8] are increasingly used frozen, with a light head. This is attractive for planetary science, where labels are scarce, but it is not obvious that features learned from web imagery carry over to rover images of Mars. Our central question is:

> *How do the features of a DINOv3 pretrained on web images extend to the Martian domain?*

We use AI4Mars [1] as one well-delimited target domain and split the question in three:

- **P1, data.** Is the Web→Mars gap a matter of image appearance? We change only how the photos look and watch the probe.
- **P2, model.** Where should the backbone be adapted? Same LoRA budget on early, middle or late blocks, compared with the probe.
- **P3, errors.** Where does the model fail and does it know? Agreement between experts, and calibration.

A preliminary *teardown* (parameters and FLOPs per block) supports these questions but is not one of them. Decision rules for every hypothesis were written before running, and all results, including inconclusive ones, are released.

2 Related Work

AI4Mars and supervised baselines. AI4Mars provides crowd-sourced terrain labels and expert-verified test sets [1]. Recent work trains CNN and transformer segmenters end to end and reports mIoU on the expert sets [3] (M1 between 0.67 and 0.70). The test-set size is not consistent across sources (323 in [1], 322 in [4], 255 in [2]); we use 322.

Label efficiency. Goh et al. [4] use SimCLR pretraining on the same test images with 1 % of the labels, but with a ResNet-50, six classes and accuracy, so numbers are not comparable to ours.

Foundation models for Mars. MANTLE [6] uses a frozen DINOv2 ViT-S/14 for HiRISE classification and rock segmentation on its own MSL data, without AI4Mars. Mars-Bench [5] benchmarks models on Mars tasks but excludes AI4Mars because of annotation errors. MarsUnet [7] adds adapters to a frozen ResNet-50 on AI4Mars with three merged classes and no absolute mIoU.

Gap. Among the 11 papers we reviewed, none evaluates DINOv3 on Mars with four-class mIoU on M1–M3, nor asks where along the depth of a frozen ViT adaptation matters on AI4Mars. We do not claim novelty for frozen backbones, ViTs or adapters on AI4Mars [6, 7, 3], and this is not an exhaustive search.

TODO: Before claiming novelty, search "LoRA by depth in ViTs" and "calibration in terrain segmentation". Read and cite DINOv3 details, DINOv2 "Rocks", ExPLoRA, GLARE.

3 Data and Protocol

Data. AI4Mars MSL NavCam images (greyscale), four classes. Training pool of 15,564 images; validation of 500 held out by acquisition time; test = three expert-labelled sets M1/M2/M3 of 322 images each (labels agreed by at least 1/2/3 experts; mapped to Swan's min1–min3 sets, an equivalence inferred from the text). Unlabelled pixels are ignored. Images are resized to 512 px, scored at label resolution.

Model. Frozen DINOv3 ViT-B/16 (LVD-1689M, 85.7 M parameters, 215 GFLOPs per 512 px image; each of the 12 blocks has 7.1 M parameters and 17.8 GFLOPs). *Probe*: linear head on the last layer (4.6 k parameters). *LoRA*: rank 8 on all linear layers of two consecutive blocks (221 k parameters), zones early (1–2), mid (6–7), late (11–12); two epochs per run, best epoch by validation.

P1 transforms. Applied to every train and test im-

Table 1: Test mIoU (4 classes), 15,564 training images. Baselines: 1 seed.

Model	Params	M1	M2
Probe, frozen DINOv3 (3 seeds)	4.6 k	0.674	0.723
DeepLabV3-R50, ImageNet	39.6 M	0.667	0.713
DeepLabV3-R50, no pretrain.	39.6 M	0.628	0.681

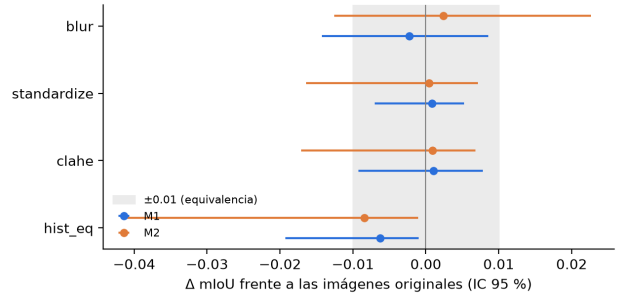


Figure 1: P1: probe mIoU under each appearance transform (M1, M2).

age: global histogram equalisation, CLAHE, per-image standardisation, Gaussian blur (σ=2 px at 512 px). A new probe is trained for each (up to 10 epochs, early stopping on validation).

Baseline. DeepLabV3-R50 trained end to end on the same images and split, pretrained on ImageNet or not, one seed, 3 epochs (70 min budget).

Statistics. Three seeds (two for LoRA, one for the baseline). Paired bootstrap of 2,000 resamples over blocks of 10 consecutive test images (neighbouring frames are similar). A difference is *real* if the 95 % CI excludes 0 and $|\Delta| \geq 0.01$ mIoU; *equivalent* if the CI lies inside $(-0.01, +0.01)$; otherwise *inconclusive*. Holm correction within each hypothesis. Hypotheses HP1, HP1b, HP2, HP3a, HP3b were registered on 02/11/2026 01:54 (internal timestamped file, not an external registry), and re-registered with the same rules before the full-data runs, *after* a pilot with 2,000 images was known; the pilot gave the same verdicts and is kept as a replicate.

4 Results

4.1 Reference point: probe vs. end-to-end

The probe matches the end-to-end model within the statistical margin (Δ M1 -0.007, CI $[-0.025, +0.008]$; without pretraining -0.046, $[-0.081, -0.016]$, real). The baseline is under-trained (3 epochs; the ImageNet run had plateaued, the random-init one was still improving) and a single seed, so the claim is "the probe matches it", not "beats it"; the gain is in cost. Published models (DeepLabV3+ R101, SegFormer-B3, UPerNet Swin-S; M1 0.67–0.70 [3]) use another split and convention, so that comparison is only indicative.

Table 2: P2: rank-8 LoRA (221 k parameters) by zone, 2 seeds, vs. probe with the same recipe.

Zone	M1	M2
Probe	0.674	0.723
Early (1–2)	0.681	0.730
Mid (6–7)	0.679	0.728
Late (11–12)	0.681	0.731

4.2 P1: is the gap about appearance?

CLAHE and standardisation are *equivalent* to the original images on M1 (+0.001, CIs inside ±0.01); histogram equalisation is slightly lower (−0.006) and blur does not sink the probe (−0.002 on M1; the largest per-class drop is soil, not sand as predicted). On M2 the intervals are wider than ±0.01, so those comparisons stay inconclusive. **HP1 (appearance does not matter): inconclusive**, although no transform gives a gain; **HP1b (blur hurts, sand most): inconclusive**, with no drop detected.

4.3 P2: where to adapt?

All three zones add 0.005–0.007 over the probe on M1 (early +0.007 $[+0.001, +0.015]$, mid +0.005 $[-0.003, +0.022]$, late +0.007 $[+0.002, +0.023]$), below the 0.01 threshold, and zones do not differ (late−early: −0.000 on M1, $[-0.006, +0.015]$; +0.001 on M2). **HP2 (late helps more than early or mid): inconclusive.** Neither the "late" prediction nor the opposite order expected from a low-level gap is supported: adaptation adds little wherever it goes. In the earlier run, partial fine-tuning of the last four blocks (28 M parameters) and a 3.5 M decoder added at most 0.012 on M1/M2, which points the same way.

4.4 P3: where does it fail, and does it know?

Label ambiguity. On the M1 images, mIoU is 0.862 on pixels also labelled in M3 (three experts agree) and 0.546 on pixels labelled only in M1; the difference is +0.315 $[+0.180, +0.413]$. **HP3a: supported.** Part of the ceiling comes from ambiguity in the labels, not from the features.

Calibration. Overall ECE is 0.029, but on pixels predicted `big_rock` the mean confidence exceeds accuracy by +0.674 $[+0.435, +0.762]$, and 45 % of `big_rock` pixels are missed. **HP3b: supported.** The probe is well calibrated on average and badly overconfident on the one class that matters for hazards.

Test-set caveat. Only 5 of the 322 M3 images contain `big_rock`, so M3 mIoU cannot rank models on that class; the M3 gap between heads (0.83 vs. 0.93) is driven by false positives.

4.5 Hypotheses and teardown

"Inconclusive" means the data neither support nor refute the hypothesis at this sample size. The earlier H1–H6 of

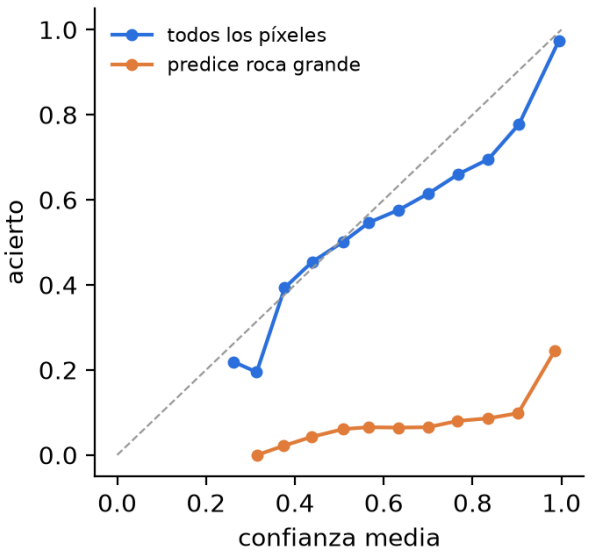


Figure 2: P3: calibration of the probe (overall and on pixels predicted `big_rock`).

Table 3: Hypotheses registered before running (full data).

Hypothesis	Verdict	Key number
HP1 appearance transforms equivalent	inconclusive	CLAHE, std.: equiv. on M1
HP1b blur hurts, sand most	inconclusive	−0.002; soil drops most
HP2 late > mid, early	inconclusive	late−early −0.000
HP3a high-agreement ≥ low by 0.05	**supported**	+0.315
HP3b calibrated, overconfident on `big_rock`	**supported**	ECE 0.029; +0.674

the layer-wise study are unchanged; their re-reading in light of v2 is post-hoc. The teardown shows that all 12 blocks cost the same (7.1 M parameters, 17.8 GFLOPs), so a depth-based adaptation budget is a choice about *where*, not about cost.

5 Limitations, Bias and Ethics

- **Single dataset and backbone.** Conclusions are limited to AI4Mars and DINOv3 ViT-B/16.
- **Statistical power.** 322 test images and effects of about 0.01 mIoU leave most hypotheses inconclusive; M2 intervals are wider than ±0.01.
- **P1.** Early stopping (patience 2) cut most probes at epoch 5, before the learning rate decayed (0.674 vs. 0.677 in the layer-wise run); the three seeds share batch order, so seed variance is understated.
- **P2.** Two seeds, two epochs per run (∼1,900 steps) and a learning rate fixed from the pilot validation, because of the time available.
- **Baseline.** One seed, 3 epochs, under-trained.
- **Labels.** Training labels are crowd-sourced and noisy; Mars-Bench excludes AI4Mars for this reason [5].
- **Safety.** `big_rock` IoU is low and the model is overconfident there: these models should not be relied on for hazard detection.
- **Data and licences.** AI4Mars is public NASA data with no personal data. DINOv3 weights are used under Meta's licence; we release code and results, not weights.
- **Process.** The first v2 pilot used 2,000 training images; this was corrected and the full-data runs are the reported ones (see the design log).

TODO: Add the authors' AI-assistance disclosure in the form the course requires.

6 Conclusion

The frozen DINOv3 features already work on Mars with almost no adaptation: a 4.6 k-parameter probe matches an end-to-end DeepLabV3 within the statistical margin. The ceiling is not set by how the image looks (P1) nor by which blocks are adapted (P2), but by label ambiguity and by the rare `big_rock` class, on which the model is overconfident (P3). Three of five hypotheses are inconclusive; the data do not separate configurations beyond about 0.01 mIoU.

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TODO: Verify full author lists, venues and page numbers of every reference against the PDFs before submission; some entries above are abbreviated.